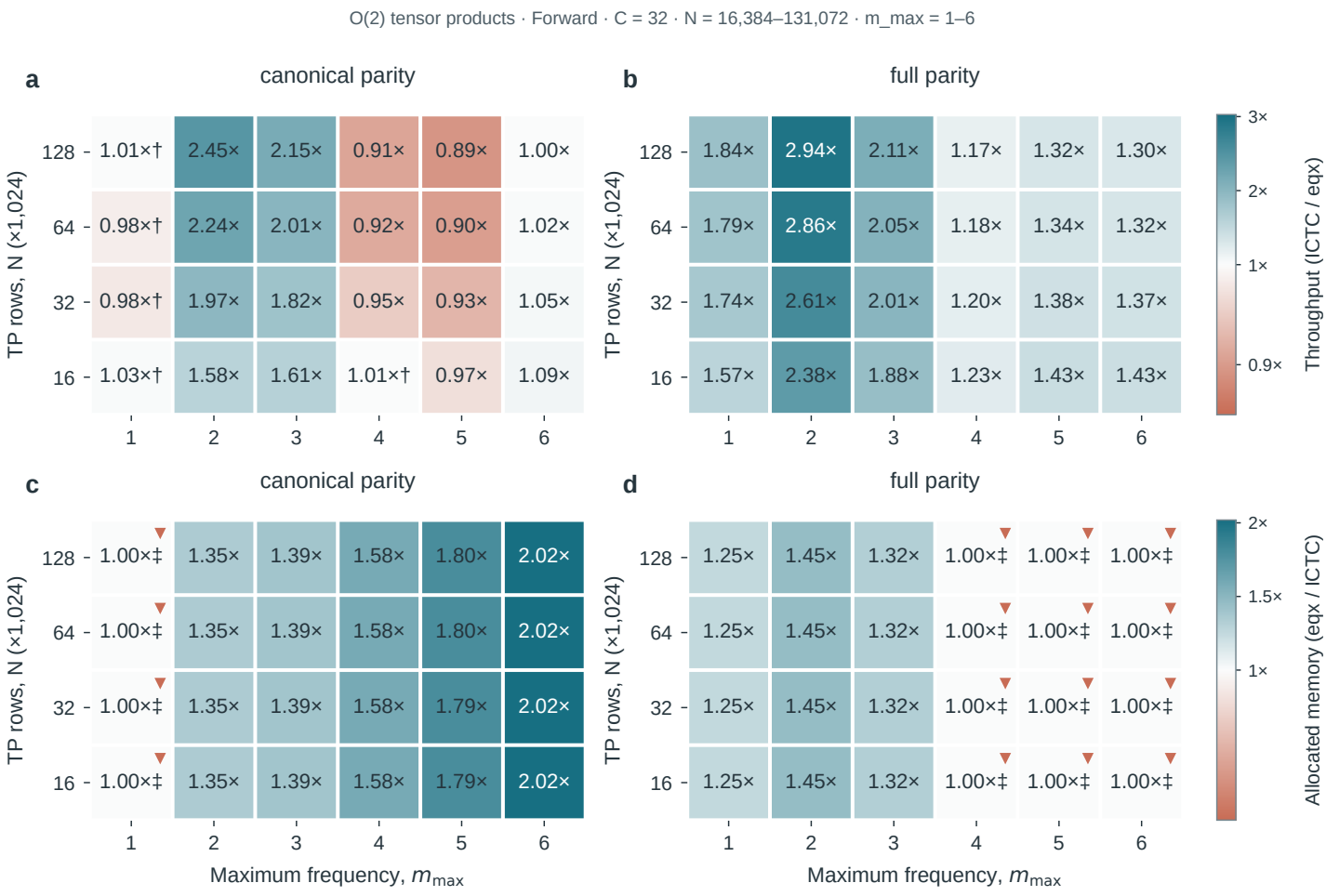

ICTC-O2 HP vs eqx: throughput and peak memory
O(2) tensor products · Forward · C = 32 · N = 16,384–131,072 · m_max = 1–6
a
canonical parity
b
full parity
c
canonical parity
d
full parity
TP rows, N (×1,024)
128
64
32
16
1.01×† 2.45× 2.15× 0.91× 0.89× 1.00×
0.98×† 2.24× 2.01× 0.92× 0.90× 1.02×
0.98×† 1.97× 1.82× 0.95× 0.93× 1.05×
1.03×† 1.58× 1.61× 1.01×† 0.97× 1.09×
1.84× 2.94× 2.11× 1.17× 1.32× 1.30×
1.79× 2.86× 2.05× 1.18× 1.34× 1.32×
1.74× 2.61× 2.01× 1.20× 1.38× 1.37×
1.57× 2.38× 1.88× 1.23× 1.43× 1.43×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.80× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.00×‡ 1.35× 1.39× 1.58× 1.79× 2.02×
1.25× 1.45× 1.32× 1.00×‡ 1.00×‡ 1.00×‡
1 2 3 4 5 6
Maximum frequency, m_{max}
Throughput (ICTC / eqx)
3×
2×
1×
0.9×
Allocated memory (eqx / ICTC)
2×
1.5×
1×
† IQR overlap * isolated timing ‡ memory increase
v0.1.5